# Competitive positioning of European semiconductor supply-chain firms in the AI era: a comparative analysis (2021–2025)

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## 1 Introduction and objective of the analysis

In recent years, the rapid adoption of artificial intelligence has increased demand for computing capacity and strengthened the role of advanced semiconductors in digital infrastructure. The development of increasingly complex processors, memories and architectures involves not only chip manufacturers but also the companies supplying the technologies required for fabrication, packaging and testing.

Several highly specialized European companies operate at different stages of this supply chain. ASML, ASM International, Besi, AIXTRON, VAT Group and Technoprobe supply technologies ranging from lithography and material deposition to advanced packaging, vacuum systems and semiconductor testing. Together, they cover several strategic stages of the semiconductor technology supply chain, forming a useful group for comparing financial performance over time.

The analysis covers 2021–2025 and examines company developments through revenue growth, research and development intensity, operating profitability, asset utilization and financial structure. The year 2020 is used as the initial reference for indexed revenue trajectories and R&D intensity. Comparisons consider both each company's development over time and the annual sample median as a common summary benchmark.

The study aims to describe changes in the financial and competitive positioning of the selected firms, identifying differences in growth, innovation, profitability and financial balance. The expansion of artificial intelligence applications provides the industrial context for this comparison; the analysis itself remains focused on financial performance observed in company financial statements.

## 2 Sample, indicators and benchmarking framework

### 2.1 Companies included in the analysis

The selection focuses on listed European companies supplying technologies for semiconductor production and testing, for which continuous and comparable financial information is available throughout the study period.

The sample includes six companies operating at strategic stages of the supply chain: ASML in lithography; ASM International and AIXTRON in deposition technologies; Besi in advanced packaging; VAT Group in vacuum systems; and Technoprobe in semiconductor testing. This composition makes it possible to examine firms exposed to distinct segments of the same European technology supply chain.

**Table 1:** Companies included in the analysis

| Company | Country | Main technology segment |
|---|---|---|
| ASML | Netherlands | Lithography |
| ASM International | Netherlands | Deposition and wafer processing |
| Besi | Netherlands | Advanced packaging |
| AIXTRON | Germany | Deposition technologies |
| VAT Group | Switzerland | High-vacuum technologies |
| Technoprobe | Italy | Semiconductor testing |

## 2.2 Data, indicators and comparison criteria

The data were obtained from the companies' consolidated financial statements and official annual reports. The principal items used are revenue, operating income, research and development expenditure, total assets, current assets and liabilities, and shareholders' equity. The specific sources associated with the raw data were recorded in the working dataset.

Five main indicators were constructed from this information: annual revenue growth, return on sales (ROS), R&D intensity, asset turnover and equity autonomy. The current ratio was also used to assess liquidity, but was not included in the subsequent principal component analysis.

Company expansion is also represented using indexed revenue, with the 2020 value set to 100. This transformation makes company trajectories comparable regardless of initial size or financial-reporting currency. R&D intensity is the ratio of research and development expenditure to revenue; ROS is operating income divided by revenue; asset turnover measures the ability to generate revenue from the asset base; equity autonomy is the share of total assets financed by equity; and the current ratio compares current assets with current liabilities.

For each indicator, the benchmark is the annual median of the six companies. In the charts, this common reference makes it possible to assess each firm's relative position alongside its time trajectory.

## 2.3 Descriptive profile of the sample

The dataset used for benchmarking and the subsequent multivariate analysis contains 30 company–year observations, representing six firms during 2021–2025, with no missing values in the indicators considered. The year 2020 is not an additional observation in the analytical dataset: it is used only as the base year for indexed revenue and as an initial reference in the R&D intensity chart.

**Table 2:** Descriptive statistics for financial indicators

| Indicator | N | Mean | Median | Std. dev. | Min | Max |
|---|---|---|---|---|---|---|
| Revenue growth (%) | 30 | 15,8 | 13,9 | 22,6 | -25,4 | 72,8 |
| R&D intensity (%) | 30 | 10,3 | 10,9 | 3,0 | 4,4 | 14,6 |
| ROS (%) | 30 | 29,1 | 28,8 | 7,3 | 12,4 | 42,4 |
| Asset turnover | 30 | 0,65 | 0,63 | 0,13 | 0,44 | 0,98 |
| Equity autonomy (%) | 30 | 64,7 | 68,7 | 18,6 | 29,1 | 88,0 |
| Current ratio | 30 | 3,72 | 3,07 | 2,12 | 1,24 | 8,06 |

The descriptive statistics show considerable variation across observations, particularly for revenue growth, financial structure and liquidity. The outlier check flags two unusually high observations: Besi's revenue growth in 2021 and VAT Group's asset turnover in 2022. Both were retained because they are consistent with the companies' financial statements. Marked differences among observations further support using the annual median rather than the mean as a benchmarking reference, because it

is less affected by extreme values.

# 3 Benchmarking evidence

The collected information is now used to compare changes in the firms' principal financial dimensions. Benchmarking considers each company's trajectory together with the annual sample median, shown by the black dashed line in the charts. The comparison covers growth and innovation, profitability and asset-use efficiency, and financial strength and liquidity.

## 3.1 Revenue expansion and innovation investment

Revenue growth is represented by an index with 2020 = 100, while innovation investment is measured by the share of revenue devoted to research and development. The two measures allow comparisons of cumulative expansion and R&D expenditure relative to turnover, respectively.

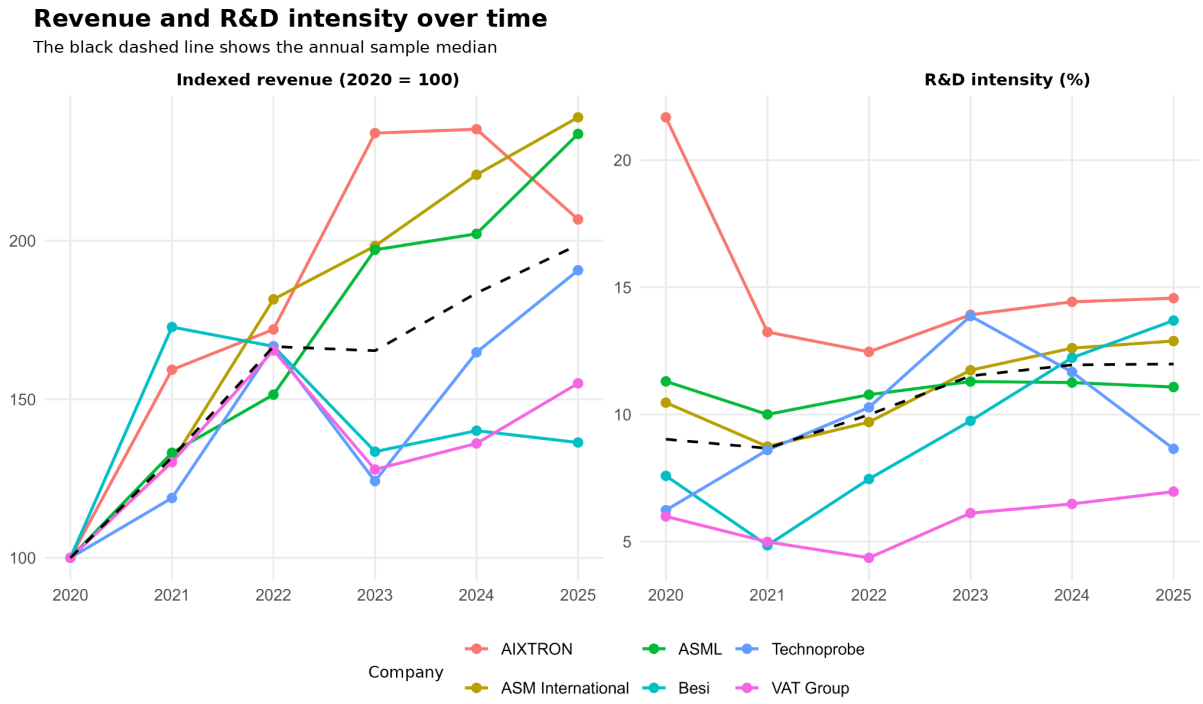



**Figure 1:** Revenue and R&D intensity over time

Figure 1 shows strong overall revenue expansion, but with different trajectories. Median indexed revenue rises from 100 in 2020 to approximately 167 in 2022, remains broadly stable in 2023, and then increases to about 199 in 2025. ASM International and ASML have the steadiest cumulative growth, reaching approximately 239 and 234, respectively, by the end of the period. AIXTRON accelerates sharply through 2023 before declining in 2025, while Besi and VAT Group show more modest growth overall. Technoprobe follows a more uneven path, contracting in 2023 and subsequently recovering.

The slowdown in 2023 occurred during a broader semiconductor-cycle correction. After record manufacturing investment in 2022, weaker chip demand and high inventory levels in the consumer and mobile segments caused a decline in spending on new facilities (SEMI, 2023). The same dynamics appear in Technoprobe's results: the company attributed its 2023 revenue decline to contraction in its reference market and high customer inventory levels (Technoprobe, 2023).

The subsequent recovery was not uniform across the supply chain. ASM International, for example, is directly exposed to deposition technologies used in the most advanced nodes: in 2024 it reported

strong demand linked to gate-all-around (GAA) architectures and high-bandwidth memory (HBM), identifying artificial intelligence as an important market driver (ASM International, 2024). Besi operates in advanced packaging, where rising demand for 2.5D solutions and data-center applications coexists with weakness in established markets such as mobile, automotive and industrial equipment (Besi, 2026). Different exposures to the technology segments driving the recovery help explain why revenue rebounded with differing timing and intensity across firms.

R&D intensity also trends upward overall. The sample median increases from approximately 9% in 2020 to about 12% in 2025. AIXTRON remains above the benchmark for most of the period, while ASM International and Besi gradually increase their R&D intensity. Technoprobe reaches its highest value in 2023 before lowering R&D expenditure as a share of revenue in subsequent years.

AIXTRON's profile illustrates that high innovation intensity does not necessarily translate into steady revenue growth. The company focuses mainly on compound semiconductors and serves power-electronics and optoelectronics markets, among others. In 2025 demand for GaN and SiC power applications remained weak, while optoelectronics benefited from rising demand for lasers and photonic solutions for high-speed data transmission (AIXTRON, 2026). The final-year revenue decline can therefore also be understood in light of differing demand trends across its end markets. Overall, the increase in median R&D intensity suggests that the cyclical slowdown did not lead to a generalized reduction in innovation investment.

### 3.2 Profitability and asset utilization

To assess whether revenue expansion was accompanied by a similar improvement in operating performance, ROS and asset turnover are examined. ROS measures the share of revenue converted into operating income; asset turnover describes the company's ability to generate revenue from its asset base.

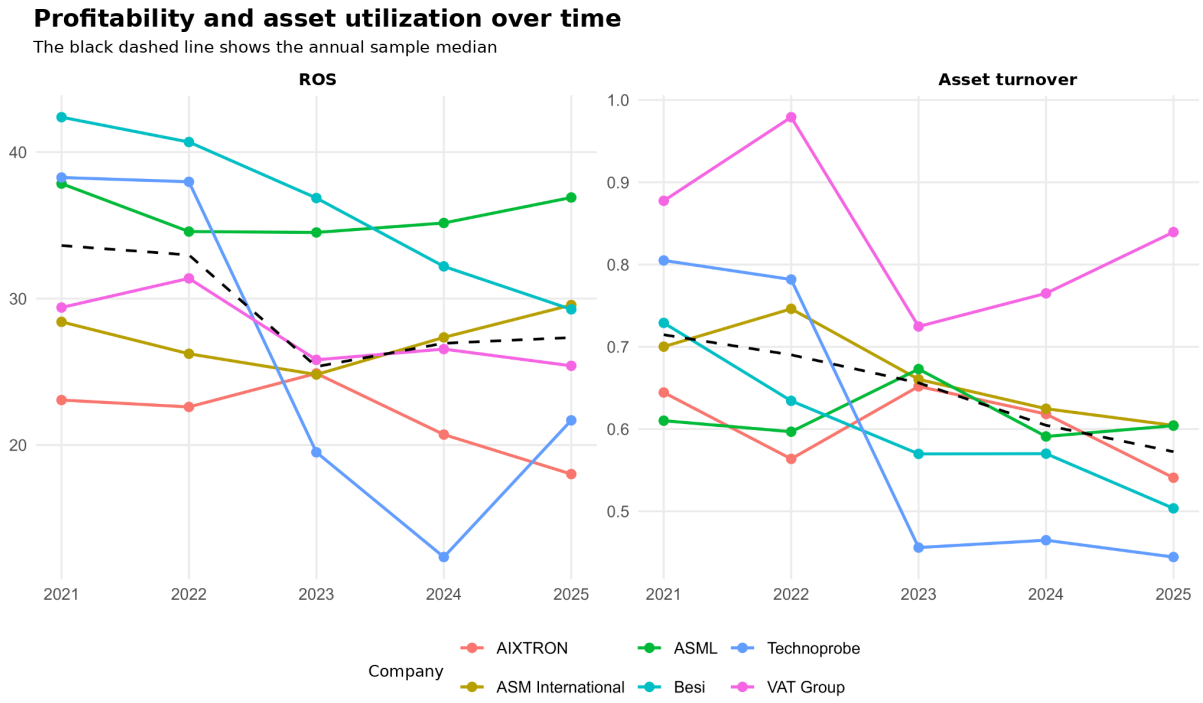



**Figure 2:** Changes in profitability and efficiency

Figure 2 shows that revenue growth does not translate into uniform improvements in profitability. Median ROS remains close to 33% in 2021–2022, falls to approximately 25% in 2023 and recovers only partially in subsequent years, reaching around 27% in 2025. ASML maintains operating profitability

above the benchmark throughout the period and remains relatively stable. Besi starts from even higher margins, but these decline progressively. Technoprobe records the sharpest change, falling from approximately 38% in 2021–2022 to about 12% in 2024, before recovering in the final year.

Asset turnover declines more broadly: the median falls from approximately 0.71 in 2021 to 0.57 in 2025. VAT Group records the highest values for almost the entire period, while Technoprobe shows a sharp decline starting in 2023. Lower turnover indicates that the asset base grew more rapidly than the revenue generated from those assets. This does not necessarily imply worsening efficiency in an absolute sense, since it may also reflect capacity expansion and investment intended to support future growth.

Technoprobe provides a clearer illustration of the connection between the business cycle, profitability and asset utilization. In 2023 weaker demand and high customer inventories caused a steep decline in sales. Lower production activity reduced the absorption of fixed costs, compressing margins; at the same time, the company chose to maintain its production structure and resources in anticipation of a subsequent recovery in volumes (Technoprobe, 2023).

During the same period, Technoprobe continued investing in production and technology and acquired Harbor Electronics in August 2023, extending its industrial footprint in the advanced printed-circuit-board segment used in probe cards (Technoprobe, 2024). The simultaneous reductions in ROS and asset turnover are therefore consistent with a period in which revenue declined while productive capacity and the asset base were largely preserved and subsequently strengthened. Revenue recovery from 2024 did not immediately restore the profitability and asset-turnover levels observed before the correction.

Overall, the comparison distinguishes revenue growth from operating performance: firms experiencing the greatest expansion are not necessarily those with the strongest improvement in margins or asset turnover. Their different paths reflect not only general market conditions but also the timing with which individual supply-chain segments move through investment, slowdown and recovery phases.

### 3.3 Financial structure and liquidity

Financial structure is examined through equity autonomy and the current ratio. Equity autonomy measures shareholders' equity as a share of total assets, while the current ratio assesses short-term financial balance by comparing current assets with current liabilities.

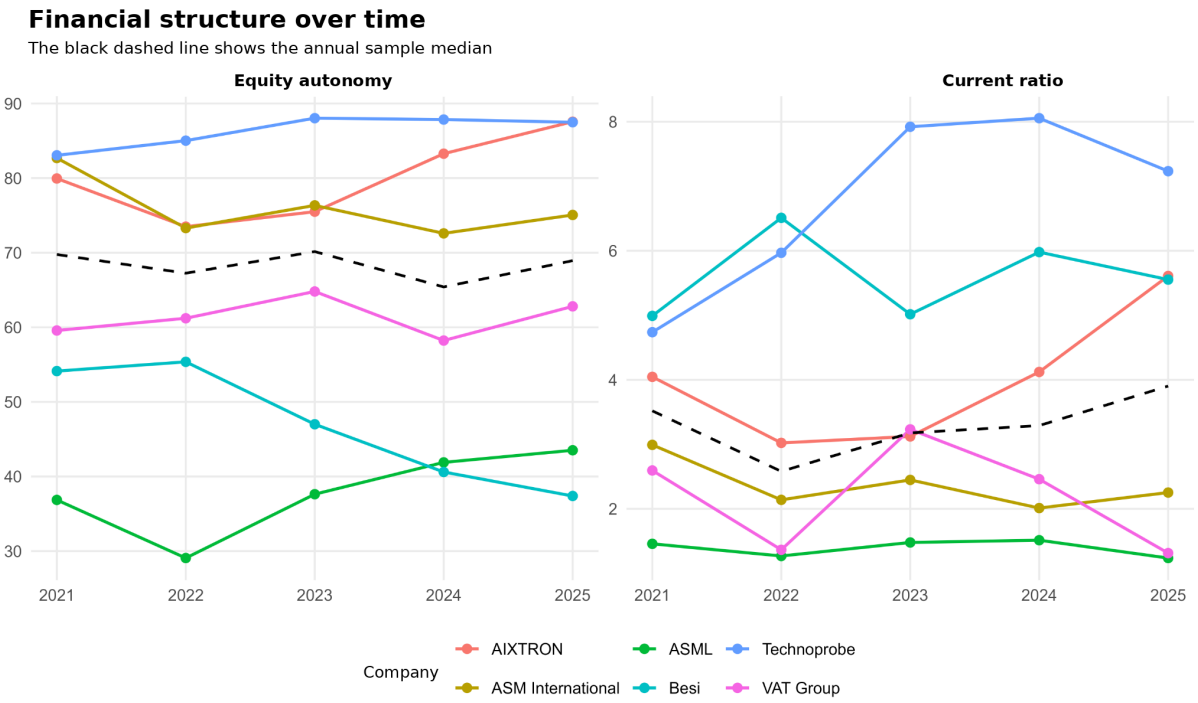



**Figure 3:** Changes in financial structure

Figure 3 highlights clearly differentiated financial profiles. Median equity autonomy remains broadly stable, fluctuating between approximately 65% and 70%. Technoprobe stays above 80% throughout the period, while AIXTRON progressively strengthens its equity autonomy to about 88% in 2025. ASM International also remains consistently above the benchmark. ASML and Besi have lower shares of equity relative to total assets, although their trajectories differ.

Liquidity differences are equally clear. The median current ratio falls from approximately 3.5 in 2021 to 2.6 in 2022, then gradually rises to about 3.9 by 2025. Technoprobe has the highest values in the later years, while Besi also shows liquidity well above the benchmark. ASML remains consistently below the sample median. The current ratio nevertheless exceeds one in every observation, indicating current assets greater than current liabilities.

The comparison therefore does not reveal one financial profile consistently associated with greater growth or profitability. ASML, for example, maintains one of the highest operating margins in the sample despite equity autonomy and liquidity below the median. Technoprobe has a particularly strong balance-sheet structure and high current ratio, but profitability falls sharply in 2023–2024. The various performance dimensions cannot be combined into a single unambiguous ranking, motivating a subsequent joint analysis through principal component analysis.

# 4 Competitive profiles and multivariate synthesis

## 4.1 PCA setup and explanatory capacity

The benchmarking results show that growth, innovation, profitability and financial balance do not necessarily move in the same direction. Principal component analysis (PCA) is applied to examine these dimensions jointly, summarizing the information in the original indicators using a smaller number of components.

The analysis uses 30 company–year observations from 2021–2025 and five variables: annual revenue growth, ROS, R&D intensity, asset turnover and equity autonomy. Before applying PCA, the indicators

are standardized to remove differences in measurement units and make their contributions to the components comparable.

**Table 3:** Eigenvalues and explained variance of the principal components

| Component | Eigenvalue | Variance (%) | Cumulative (%) |
|---|---|---|---|
| PC1 | 2,309 | 46,2 | 46,2 |
| PC2 | 1,324 | 26,5 | 72,6 |
| PC3 | 0,784 | 15,7 | 88,3 |
| PC4 | 0,312 | 6,2 | 94,6 |
| PC5 | 0,271 | 5,4 | 100,0 |

The first two components have eigenvalues above one and together explain 72,6% of total variance. The two-dimensional PC1–PC2 representation therefore preserves a large share of the information contained in the five indicators and is used to summarize firms' financial profiles.

## 4.2 Interpretation of the components

To assign an economic interpretation to the two retained dimensions, associations between the original indicators and the principal components are examined.

**Table 4:** Associations between original variables and PC1/PC2

| Variable | PC1 | PC2 |
|---|---|---|
| Revenue growth (ΔRIC) | -0,471 | 0,481 |
| ROS | -0,765 | -0,472 |
| R&D intensity | 0,809 | -0,227 |
| Asset Turnover (AT) | -0,778 | 0,438 |
| Equity autonomy (AF) | 0,492 | 0,792 |

The first component, accounting for 46,2% of variance, mainly contrasts R&D intensity and, to a lesser extent, equity autonomy with operating profitability and the ability to generate revenue from assets. Positive PC1 values are therefore associated with greater R&D intensity, while ROS and asset turnover show strong negative associations.

The second component explains 26,5% of variance and is most strongly associated with equity autonomy, followed by revenue growth and asset turnover; ROS lies primarily on the opposite side. PC2 can thus be interpreted mainly as a dimension of balance-sheet strength and growth dynamics. Component signs are conventional: the interpretation concerns contrasts among variables, not an automatic ranking from better to worse.

## 4.3 Firm positions and trajectories

The factor plane allows company positions and their changes over time to be examined simultaneously. Trajectories are interpreted by linking movements in the plane to the benchmarking results, without treating the components as a ranking.

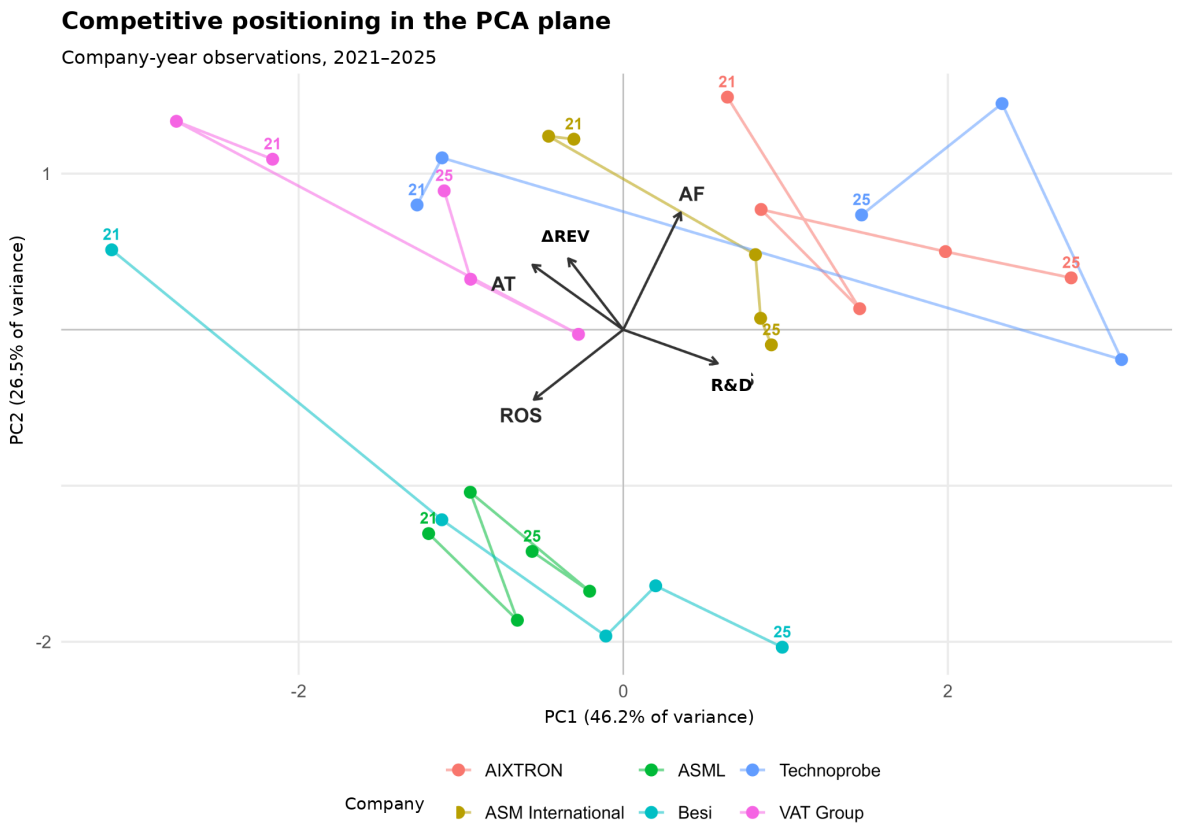



**Figure 4:** Competitive positioning of the firms in the PCA plane

Figure 4 shows how the differences identified by benchmarking are reflected in the companies' joint positions. The origin represents the mean profile of all standardized observations in the full sample and is not the annual median benchmark used in the previous charts.

**AIXTRON and ASML.** AIXTRON remains on the positive side of the first component throughout the period and moves farther right in 2025. This position is consistent with high R&D intensity and strong equity autonomy, combined with lower profitability and asset turnover in the more recent years. ASML has an almost opposite profile and remains in the lower-left quadrant: its location is associated particularly with persistently high ROS, while lower equity autonomy contributes to its negative PC2 scores. The comparison therefore identifies two different competitive configurations, one characterized mainly by innovation intensity and balance-sheet strength, and the other by operating profitability.

**Besi.** Besi shows one of the largest movements in the factor plane. Between 2021 and 2025, it shifts from the upper-left to the lower-right quadrant, with PC1 moving from strongly negative to positive values and PC2 moving from positive to strongly negative values. The PC1 shift is consistent with the progressive increase in R&D intensity and the reductions in ROS and asset turnover; movement toward lower PC2 values mainly reflects slower growth and declining equity autonomy. The trajectory therefore describes a transformation in the firm's profile, rather than a simple overall improvement or deterioration.

**Technoprobe.** Technoprobe follows an equally dynamic path. In 2021–2022 it occupies the upper-left part of the plane, consistent with high ROS and asset turnover and strong equity autonomy. Its profile changes markedly in 2023: revenue growth becomes negative, ROS falls from approximately 38% to 19,5%, and asset turnover drops from 0,78 to about 0,46, while R&D intensity reaches its highest value in the period. Profitability declines further in 2024; in 2025, the trajectory moves partly back

toward the center, as ROS recovers and the balance-sheet structure remains very strong. This pattern highlights an episode of pronounced discontinuity followed by signs of rebalancing in the final year.

Overall, PCA confirms that the firms cannot be placed in a single unambiguous ranking. Their positions depend on the combination of growth, innovation, profitability, asset utilization and financial strength; companies in different parts of the plane may therefore have strengths in different dimensions.

# 5 Conclusions

During 2021–2025, the sample shows strong overall expansion, interrupted by the 2023 cyclical correction and followed by a selective recovery. Revenue growth is accompanied by generally rising R&D intensity, but its effects on profitability and asset utilization are not uniform: ROS recovers only partially after 2023, while asset turnover declines progressively.

Differences among the firms are therefore substantial. ASML and ASM International have the steadiest growth trajectories; AIXTRON combines high innovation intensity with strong equity autonomy; and Besi and Technoprobe experience the greatest changes in their profiles. PCA, whose first two components summarize 72,6% of variation, confirms that there is no single model of success: profitability, innovation, growth and financial strength occur in different combinations.

Against the backdrop of increasing demand for computing capacity and AI applications, the principal result is the selective nature of supply-chain development. The benefits of recovery emerge at different times and with different intensities, depending on technologies and end markets. The competitive position of the firms examined thus depends on their ability to combine technological specialization, innovation and financial sustainability across different phases of the semiconductor cycle.

# References


SEMI (2023), *2024 Global Fab Equipment Spending Recovery Expected After 2023 Slowdown, SEMI Reports*. Available at: https://www.semi.org/en/news-media-press-releases/semi-press-releases/2024-globaL-fab-equipment-spending-recovery-expected-after-2023-slowdown-semi-reports.

Technoprobe (2023), *Interim Condensed Consolidated Financial Statements as at 30 June 2023*. Available at: https://www.technoprobe.com/wp-content/uploads/2023/08/ENG-Fascicolo-Technoprobe-SpA-30.06.2023.pdf.

ASM International (2024), *Q3 2024 Investor Presentation*. Available at: https://www.asm.com/downloads/presentations-other/2024-q3-investor-presentation.

BE Semiconductor Industries (2026), *Q4-25 and Full Year 2025 Results*. Available at: https://www.besi.com/investor-relations/press-releases/details/be-semiconductor-industries-nv-announces-q4-25-and-full-year-2025-results/.

AIXTRON (2026), *AIXTRON with robust results in a soft market environment*. Available at: https://www.aixtron.com/en/investors/AIXTRON%20with%20robust%20results%20in%20a%20soft%20market%20environment_n13896.

Technoprobe (2024), *Annual Financial Report 2023*. Available at: https://www.technoprobe.com/wp-content/uploads/2024/03/xTechnoprobe-SpA-ENG_RELAZIONE-ANNUALE-2023-FASCICOLO-COMPLETO-1.pdf.